

NENJIANG, China

WMO: 505570

Lat: 49.1561N

Lon: 125.2278E

Elev: 797

StdP: 14.28

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-26.7	-36.7	0.7	-30.6	-32.6	0.9	-26.4	18.2	0.8	16.2	1.9	1.5	70	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	88.1	67.2	84.7	66.9	81.8	66.7	73.4	80.1	71.7	78.7	70.0	77.7	11.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	119.6	76.7	69.5	111.8	75.4	67.5	104.2	73.6	37.7	80.3	36.1	78.3	34.6	77.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	23.3	19.6	16.7	DB	-33.7	94.4	6.0	3.6	-38.0	97.1	-41.5	99.2	-44.9	101.2	-49.2	103.9
(5)				WB	-33.9	76.5	6.0	2.2	-38.1	78.1	-41.6	79.3	-45.0	80.6	-49.3	82.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	-8.7	-0.1	18.4	39.8	55.1	67.1	71.0	66.5	54.7	37.7	13.6	-5.0
	DBStd	29.61	9.25	9.53	12.16	8.51	7.95	6.49	4.68	5.21	7.38	8.71	12.41	8.75
	HDD50	7784	1821	1404	978	323	35	0	0	0	36	391	1091	1705
	HDD65	11607	2286	1824	1443	756	323	51	7	43	316	847	1541	2170
	CDD50	2073	0	0	0	18	194	512	652	511	177	9	0	0
	CDD65	419	0	0	0	0	17	112	194	89	7	0	0	0
	CDH74	3724	0	0	0	13	264	1193	1477	681	95	1	0	0
	CDH80	1060	0	0	0	2	71	421	425	126	15	0	0	0
Wind	WSAvg	6.7	4.1	4.9	7.4	9.5	9.8	7.6	6.6	6.2	6.9	7.3	6.1	4.4
Precipitation	PrecAvg	18.9	0.1	0.1	0.2	0.8	1.4	2.8	5.6	4.4	2.4	0.7	0.2	0.2
	PrecMax	27.9	0.4	0.6	0.6	3.1	2.9	6.1	15.3	10.4	4.3	1.9	0.8	0.6
	PrecMin	13.3	0.0	0.0	0.0	0.0	0.1	0.7	1.2	0.7	0.3	0.0	0.0	0.0
	PrecStd	3.5	0.1	0.1	0.2	0.7	0.7	1.4	2.6	2.4	1.2	0.5	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	33.9	52.8	75.2	87.0	94.5	92.3	87.0	81.8	68.4	46.4	22.4
		MCWB	15.4	27.7	38.9	51.7	59.2	67.1	70.3	69.6	64.5	52.9	38.2	19.7
	2%	DB	12.9	26.4	46.2	67.3	81.3	88.6	88.2	83.3	76.4	61.3	40.0	16.6
		MCWB	10.6	22.7	35.8	47.4	57.3	65.3	70.4	70.1	61.6	48.5	33.1	14.3
	5%	DB	9.1	20.6	41.0	61.7	76.0	84.9	84.5	80.6	72.7	56.4	34.7	12.1
		MCWB	7.2	17.5	32.5	44.8	54.9	64.3	69.7	68.7	59.6	45.1	29.7	10.3
	10%	DB	5.6	16.0	36.7	56.3	71.1	81.0	81.6	77.9	69.0	51.9	30.6	8.6
		MCWB	4.0	13.7	30.3	42.1	52.9	63.0	69.0	67.4	57.7	42.7	26.9	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	27.4	41.4	53.6	61.9	71.4	76.5	75.1	68.5	55.3	38.5	20.2
		MCDB	17.9	32.9	50.6	71.1	76.9	84.5	84.2	81.1	74.8	65.9	45.5	22.4
	2%	WB	10.7	22.6	36.3	48.8	59.2	69.0	74.2	72.8	64.9	49.9	33.4	14.4
		MCDB	12.9	26.5	44.8	64.1	75.6	82.1	81.2	78.5	72.1	58.4	39.3	16.5
	5%	WB	7.1	17.6	33.3	46.0	57.1	67.1	72.6	71.1	62.1	46.5	30.2	10.4
		MCDB	9.0	20.5	40.2	59.7	72.1	78.1	79.6	77.4	69.1	54.1	34.2	12.1
	10%	WB	4.1	13.6	30.5	43.1	54.8	65.3	71.0	69.2	59.5	43.5	27.1	7.0
		MCDB	5.5	15.9	36.5	55.0	67.5	76.1	78.1	75.8	66.9	51.1	30.5	8.5
Mean Daily Temperature Range	5% DB	MDBR	19.8	23.9	22.4	22.6	22.7	20.4	17.4	18.8	22.3	19.8	18.0	18.0
		MCDBR	21.3	25.8	24.3	31.4	29.8	27.1	22.8	23.2	27.4	26.8	20.9	20.0
		MCWBR	19.4	22.4	16.2	17.4	13.4	10.5	9.3	11.5	14.1	16.0	15.6	18.3
	5% WB	MCDBR	21.1	25.7	22.6	27.8	24.6	21.2	16.9	18.1	21.4	23.5	19.8	19.8
		MCWBR	19.2	22.3	15.4	15.9	12.0	9.2	9.1	10.2	12.2	15.3	15.2	18.1
Clear-Sky Solar Irradiance	taub	0.276	0.301	0.356	0.455	0.529	0.514	0.491	0.440	0.394	0.371	0.309	0.284	
	taud	2.197	2.135	2.059	1.941	1.818	1.967	2.074	2.191	2.253	2.191	2.245	2.171	
	Ebn at Noon	243	264	266	249	236	241	245	252	253	233	227	219	
	Edh at Noon	24	33	43	54	64	56	49	42	36	32	23	22	
All-Sky Solar Radiation	RadAvg	576	961	1411	1620	1790	1886	1733	1571	1292	892	548	433	
	RadStd	26	44	86	131	134	208	167	129	101	51	42	35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page